

Task T1 Report — Project 3: Graph Anomalies

CS60078 Complex Network Theory | Spring 2025–26
Group-3: 23EX10026, 22CS30072, 22IM30034

Paper: GAD-NR: Graph Anomaly Detection via Neighborhood Reconstruction

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Venue: WSDM 2024

1. Problem Statement

Detecting anomalous nodes in graphs is a fundamental task in network analysis, with real-world relevance in fraud detection, spam identification on social platforms, network intrusion detection, and financial transaction monitoring. The difficulty arises because graph data is inherently multi-modal — each node carries its own feature vector, and the topology of the graph encodes additional structural context. Anomalies can therefore manifest in different ways depending on which of these modalities is affected.

The paper identifies three distinct categories of graph anomalies. **Contextual anomalies** are nodes whose feature vectors differ markedly from their local neighbourhood — e.g., a fake account whose textual content is inconsistent with the accounts it is connected to. **Structural anomalies** are nodes embedded in unusually dense clique-like subgraphs, a pattern associated with coordinated fake-review rings or bot clusters on social media. **Joint-type anomalies** are nodes that can only be flagged by considering both features and connectivity simultaneously — for instance, a phishing email sender that connects across many disparate communities while having atypical communication patterns.

The central problem the paper addresses is: *can a single, unified model reliably detect all three anomaly types on real attributed graphs, without requiring labelled training data?* This is a non-trivial goal because the three types place conflicting demands on any detection algorithm.

2. Previous Approaches and Their Limitations

Prior work on graph anomaly detection can be broadly bucketed into three families, each with characteristic blind spots.

Structure-only methods such as SCAN and centrality-based approaches work purely with the graph topology. They are reasonably effective at finding structural anomalies — dense subgraphs stand out in degree distributions and clustering coefficients — but they are completely blind to contextual anomalies because they never look at node features at all. False-alarm rates are also high for joint-type anomalies.

Feature-only methods like LOF, Isolation Forest, and MLPAE work in the opposite direction: they treat each node's feature vector in isolation and apply classical outlier detection on top of it. While these methods handle contextual anomalies reasonably well, they completely ignore graph connectivity, so structural and joint-type anomalies go undetected.

Graph Autoencoder (GAE) based methods — DOMINANT, GCNAE, AnomalyDAE — combine GNN encoders with decoders that try to reconstruct the original adjacency structure. The intuition is that normal nodes will have low reconstruction error and anomalies will have high error. This works well for contextual and densely-connected structural anomalies. However, the reconstruction objective is a *proximity-driven link prediction* task: the decoder only checks whether edges exist between pairs of nodes. This pushes connected nodes together in latent space but tells the model nothing about the broader neighbourhood context of a node. As a result, joint-type anomalies — whose oddity lies precisely in the combination of connectivity pattern and feature distribution — are largely missed. There is also a non-trivial aggregation problem: to assign an anomaly score to a node from edge-level reconstruction errors, one must decide how to combine losses across all incident edges, and there is no principled way to do this.

3. The Proposed Solution: GAD-NR

GAD-NR (Graph Anomaly Detection via Neighborhood Reconstruction) takes a different angle. Instead of asking whether individual edges can be predicted, it asks whether the *entire one-hop neighbourhood* of a node can be reconstructed from that node's compressed representation. This is a richer target, and the authors argue — convincingly, given the experimental results — that it naturally captures all three anomaly types.

Encoder. A standard GNN (GCN or GraphSAGE) takes the input graph and produces a hidden representation h_u^L for each node u . If node features are very high-dimensional and sparse (e.g., bag-of-words), a random linear projection is first applied to get a dense initial embedding.

Decoder — three reconstruction targets. From $h_u^{(1)}$, the decoder predicts three things for every node u :

(i) **Self-feature reconstruction:** An MLP predicts the original node feature vector. The loss is the L2 distance between prediction and ground truth. This term is most sensitive to contextual anomalies.

(ii) **Node degree reconstruction:** A separate MLP predicts the degree d_u . The L2 loss on degree is especially useful for structural anomalies that form unusually dense cliques.

(iii) **Neighbour feature distribution reconstruction:** Rather than trying to reconstruct each neighbour individually (which would require solving a costly assignment problem), the paper approximates the distribution of neighbour features as a multivariate Gaussian $N(\mu_u, \Sigma_u)$, where μ and Σ are empirical mean and covariance computed from actual neighbours. The decoder predicts a second Gaussian $N(\hat{\mu}_u, \hat{\Sigma}_u)$ via MLPs, and the reconstruction loss is the closed-form KL divergence between the two Gaussians. This term drives detection of joint-type anomalies.

The overall loss and anomaly score are weighted sums: $L'_u = \lambda_x * L_u^x + \lambda_d * L_u^d + \lambda_n * L_u^n$. Tuning these weights allows practitioners to emphasise whichever anomaly type is most relevant in a given application, though the paper shows that a single fixed configuration already outperforms all baselines across six datasets.

Key technical improvement over NWR-GAE. An earlier model, NWR-GAE, also used neighbourhood reconstruction but computed the loss via the Wasserstein distance with Hungarian matching, which has $O(d^3)$ per-node complexity. The Gaussian approximation reduces this to $O(d)$ and, crucially, prevents overfitting to anomalous behaviours — NWR-GAE's expressive empirical-distribution matching actually hurts anomaly detection because it fits anomalous data too well.

How the limitations are addressed. Because all three losses operate at the node level rather than the edge level, there is no aggregation ambiguity. Feature-only models are subsumed by the self-feature term. Structure-only models are subsumed by the degree term. The neighbourhood distribution term adds the cross-modal signal that prior GAE models lacked, enabling joint-type anomaly detection for the first time in a unified framework.

4. Datasets

Experiments are run on six real-world attributed graphs. The table below summarises their key statistics. For the Cora dataset, no ground-truth anomaly labels exist, so synthetic contextual and structural anomalies are injected following the BOND benchmark protocol.

Dataset	Nodes	Edges	Features	Avg. Deg.	Anomaly %
Cora	2,708	11,060	1,433	4.1	5.1% (injected)
Weibo	8,405	407,963	400	48.5	10.3%
Reddit	10,984	168,016	64	15.3	3.3%
Disney	124	335	28	2.7	4.8%

Books	1,418	3,695	21	2.6	2.0%
Enron	13,533	176,987	18	13.1	0.4%

Table 1: Summary statistics of the six benchmark datasets.

To give a concrete sense of the data: **Cora** is a paper-citation network where each node is a research paper represented by a 1,433-dimensional bag-of-words vector from its content, and each directed edge indicates that one paper cites another. The graph is sparse with an average degree of 4.1. Injected structural anomalies are created by picking 70 random nodes and forcing 10 of them into fully-connected cliques; injected contextual anomalies replace a node's feature vector with the most distant feature vector found among a random candidate pool. **Enron** is an email-interaction network (nodes = email addresses, edges = communication events) where spam senders are labelled as anomalies; the anomaly rate is very low (0.4%), making it a challenging imbalanced-detection problem. The small-scale **Disney** dataset is a co-purchase network of movie titles where anomaly labels were obtained by majority voting from students.

5. Results and Conclusion

The primary evaluation metric is the area under the ROC curve (AUC). An AUC of 1.0 indicates perfect detection; 0.5 indicates random guessing. The table below shows benchmark detection results (averaged over 5 runs). GAD-NR uses a fixed hyperparameter setting ($\lambda_x = 0.8$, $\lambda_d = 0.5$, $\lambda_n = 0.001$) throughout — no per-dataset tuning is needed to outperform the baselines.

Method	Cora	Weibo	Reddit	Disney	Books	Enron
DOMINANT	82.7	85.0	56.0	47.1	50.1	73.1
AnomalyDAE	83.4	91.5	55.7	48.8	62.2	54.3
CONAD	78.8	85.4	56.1	48.0	52.2	71.9
GAD-NR	87.6	87.7	58.0	76.8	65.7	80.9

Table 2: Benchmark AUC scores (avg. over 5 runs, fixed hyperparameters). Best scores per dataset in bold in original paper; GAD-NR row highlighted.

GAD-NR achieves the best or second-best score on five of the six datasets. The gains are particularly striking on Disney (+28 points over DOMINANT) and Enron (+7 points). On Weibo, where spectral methods like Radar achieve 98.9 AUC, the dataset characteristics seem to favour a different kind of detection, and GAD-NR scores slightly lower — but still competitive at 87.7.

Ablation experiments confirm that each component of the decoder carries weight. Removing the neighbour reconstruction term ($\lambda_n = 0$) causes the sharpest overall drop in performance. Removing the self-feature term ($\lambda_x = 0$) hurts contextual anomaly detection disproportionately. The degree term contributes most when paired with the neighbour term for joint-type anomalies.

On running time, GAD-NR is dramatically faster than NWR-GAE: per-epoch time drops from ~51 seconds to ~2.35 seconds on Cora, and from ~65 seconds to ~0.01 seconds on Enron. This is a direct consequence of replacing $O(d^3)$ Hungarian matching with $O(d)$ KL-divergence computation.

Conclusion. GAD-NR is a clean and principled solution to a long-standing limitation of graph anomaly detection: the inability to handle all three anomaly types simultaneously. The key insight is to reconstruct the neighbourhood rather than individual edges, and to approximate the neighbourhood's feature distribution as a Gaussian to keep things tractable. The model is robust to hyperparameter choice, scales well to large graphs, and outperforms a wide array of baselines across diverse real-world datasets. Future directions the paper suggests include extending neighbourhood reconstruction to multi-hop settings and exploring its utility for dynamic or heterogeneous graphs.